

Uncertainty: how much to trust a sensor

M3 • 29 Nov 2026 - 27 Dec 2026 • 75 hours

This month's deliverable

In M02, you obtained an angle, force or position as a number. M03 adds variability, error sources and conditions under which a conclusion remains valid. Deliver a **portfolio of uncertainty and optimisation work**: a 30-repeat measurement experiment, encoder and IMU noise models, calibration through optimisation, poor-conditioning diagnostics and a report on link-endpoint position uncertainty.

This month's questions are practical: does encoder jitter differ from actual motion; how should two angle estimators be compared; how much independent information does a long recording contain; why did an optimiser produce a plausible but incorrect scale; how far does joint error move a foot or fingertip? Do not yet build a Kalman filter, machine-learning model or QP controller: first learn to check the assumptions they will require.

Shared object and data

Continue with the M02 link $L=0.12$ m, $p(q)=(L \cos q, L \sin q)$ and calibration $q=a \cdot u+b$. Create data/synthetic, data/measured, experiments, calibration, uncertainty and report. In the data manifest, record each file's source, units, random-number generator seed, date, independent-run identifier and how ground truth was obtained. Keep synthetic ground truth separate from algorithm inputs.

The physical experiment needs no actuator: place a stationary IMU in the same fixture thirty times. An existing phone that exports raw readings also works. If no sensor is available, first practise the procedure with synthetic data; the real-measurement section remains incomplete until the experiment is done. Generated noise does not replace measured device repeatability.

Prerequisites and scope

You need array means, matrices, least squares, derivatives and the meaning of the Jacobian from M02. In the first two hours, explain the differences between true value, measurement, model residual and numerical error. Review why the calculated 'velocity' of a stationary encoder can fluctuate. M03 adds ways to describe these fluctuations and assess the robustness of conclusions.

Begin with definitions: a random variable models possible outcomes; a sample is the data actually obtained; a parameter is an unknown distribution characteristic; an estimator is a rule for computing from a sample. One distribution plot does not prove normality, and optimiser success does not establish physically correct calibration. Both distinctions must appear in your checks.

75 hours: experiments, analysis and reproducibility

Four work blocks

A, hours 1-20: 10 h theory + 10 h practice. Random variables, distributions, expectation, variance, covariance, correlation, independence, conditional probability and Bayes' rule. Practical deliverables: encoder quantisation and IMU noise models, and diagnosis involving a rare event.

B, hours 21-40: 10+10 h. Sampling plans, confidence intervals, effect size, outliers, robust summaries, hypotheses, repeatability and data leakage. Deliverables: a 30-repeat protocol, raw recordings and a comparison of two estimators across independent runs.

C, hours 41-60: 10+10 h. Objective functions, constraints, gradients, Hessian meaning, gradient descent, line search, local minima, least squares and regularisation. Deliverables: calibration with `scipy.optimize`, held-out validation and residual analysis.

D, hours 61-75: 5+10 h. SVD, pseudoinverse, poor conditioning, covariance propagation through a Jacobian, Monte Carlo and bootstrap. Deliverable: an endpoint uncertainty region and checks of linear-approximation limits. Total: 35 h of theory and analysis plus 40 h of practice.

Organising short and long sessions

Divide each 10 h of theory into five two-hour sessions and the final 5 h into 2+2+1. Each 2 h session should cover one concept, one hand-worked example and one decision about the forthcoming experiment. Schedule 10 h practical blocks in available Sunday or holiday sessions from the original plan; divide each into five 2 h segments and count breaks separately.

The calendar window follows the original plan. A boundary date may coincide with a neighbouring module; holidays, rest and integration stages are preserved. Allocate step hours sequentially to available sessions, without double counting or adding another 'compensation' hour.

Experiment log

Before collecting data, record the question, primary metric, unit of independent repetition, recording-exclusion criterion and stopping rule. Afterwards, add actual deviations from the protocol. Do not delete an inconvenient recording because it worsens the mean: retain it with a reason and, if necessary, present analysis with and without exclusion.

If time is short, remove optional CVXPY comparison and complex interactive plots. Retain 30 repeats, basic calibration, splitting data by run, SVD diagnostics and Monte Carlo calculation. Another book cannot replace the measurement work. Being able to say 'these data do not yet establish the claim' is a valid laboratory outcome.

Probability and sensor-noise models

Topics to study

Study discrete and continuous random variables, interval probabilities, probability density and cumulative distribution functions. For sensors, use a normal distribution as a working model of small disturbances, a uniform distribution as a quantisation simplification, and a Bernoulli distribution for 'packet lost'. Expectation describes the centre; variance is the mean squared deviation; standard deviation has the same units as the measurement.

Covariance describes joint variation of two channels; correlation normalises it by their standard deviations. Zero correlation alone does not establish independence. Consecutive stationary-IMU samples can be dependent because of internal filters and drift. A thousand samples therefore need not equal a thousand independent experiment repeats.

Task 1. Encoder quantisation

Input: true angle q within $-0.5 \dots 0.5$ rad and quantisation step $\Delta = 0.001$ rad. Compute rounded q_{meas} , error $e = q_{\text{meas}} - q$, a histogram and empirical cumulative distribution function. For uniform coverage of quantisation intervals, compare variance with $\Delta^2/12$. Then take a stationary angle exactly between two codes and explain why this distribution model need no longer describe the observations.

Deliverables: generator, two different input trajectories and a table of mean, standard deviation (SD) and maximum error. Check $|e| \leq \Delta/2$ under the chosen rounding convention, apart from stated numerical tolerances. Failure case: the model saturates beyond range; an error does not become uniform merely because the data came from an encoder.

Task 2. Shared noise on two IMU axes

Generate 5000 synthetic error pairs by adding each axis's independent noise to a shared component. Record the covariance matrix and explain its units: for acceleration, $(\text{m/s}^2)^2$. Plot the point cloud and compare it with independent data having the same per-channel variance. Equal axis standard deviations do not imply equal uncertainty in their sum.

Check covariance symmetry and nonnegative eigenvalues, allowing for rounding. Add slow shared drift and compare the recording's first and second halves. Deliver three point clouds and explain the effect shown by each. The aim is to recognise when constant-noise assumptions fail, rather than force a 'correct' distribution onto the data.

Conditional probability and diagnostic reliability

Bayes' rule for a clear robot event

Study $P(A|B)$, the law of total probability and Bayes' rule using a frequency table. Example: A is a genuine foot-contact failure; B is a threshold diagnostic trigger. Sensitivity $P(B|A)$ and the probability of a real failure after a trigger $P(A|B)$ answer different questions. The latter also depends on event prevalence.

For the teaching example, use failure probability 1%, sensitivity 95% and a 5% false-trigger probability without failure. These are illustrative calculation values, not properties of an actual sensor. Among 10 000 virtual episodes, expect 95 true and 495 false triggers; about 16.1% of all alerts are true. Check using a probability tree or table.

Task 3. Rare events and thresholds

Create an episode generator with a known label and diagnostic feature. Calculate TP, FP, FN, TN, then $\text{precision} = TP/(TP+FP)$ and $\text{recall} = TP/(TP+FN)$. Show how both metrics and trigger frequency change across several thresholds. If a denominator is zero, return 'undefined' rather than silently assign a perfect score.

Repeat with event probabilities 0.1%, 1% and 10%, retaining the feature's conditional distributions. Deliver a table and short note identifying the more consequential error type for this test bench. Here, diagnostics only create log events; this model does not verify a robot's actual protective functions.

Task 4. Why independence needs checking

Generate white noise and a correlated series $e_t = 0.95 \cdot e_{t-1} + \eta_t$. Scale them to approximately equal SD; compare time plots and adjacent-sample correlation. Average over equal-length windows and measure variation of means across non-overlapping windows. Equal sample counts do not guarantee equal improvement.

Control check: shuffle the correlated series and repeat. Shuffling preserves the marginal distribution but destroys temporal structure. Save two plots and explain why randomly splitting rows of one recording can yield an overly optimistic algorithm evaluation. Exact effective-sample-size theory is optional; the meaning of an independent run must be understood now.

Statistics: a real 30-repeat experiment

Write the protocol before recording

Task 5 assesses the repeatability of a stationary IMU repositioned in the same fixture. Choose a device that exports timestamps and raw accelerometer and gyroscope readings; record its settings and available rate. Perform 30 separate placements, keeping the device still each time, waiting the same predetermined settling interval and recording for the same duration. Choose a duration supported by export; the teaching guideline is 2 s after settling.

The repeat unit is one independent run after repositioning, not one CSV row. Record run_id, fixture orientation, time, temperature if available, observed interference and setting changes. Within each run, calculate the axis mean and SD; across 30 runs, examine variation of these means.

Repositioning adds geometric uncertainty, so describe the result as repeatability of the whole procedure.

Calculations and interpretation

For the 30 means, find the mean, median, SD and interquartile range. If assumptions are reasonable, construct a 95% t-confidence interval for the mean: $\text{mean} \pm t \cdot s / \sqrt{n}$. This interval concerns the procedure's mean; it is neither a range covering 95% of future individual readings nor a promise to cover a particular sensor's true angle.

If run independence or stability fails, show a time plot and limit the conclusion. Do not treat 30 as universally sufficient: justify this pilot sample size using available time and desired interval width, then estimate the additional independent repeats needed to halve the interval width under the same assumptions.

Task 6. Outlier robustness and effect size

Compare two constant-level estimates on the same 30 runs: within-recording arithmetic mean and median. Add labelled individual outliers to copies of several recordings. Calculate the change in error relative to the clean copy and paired estimate differences, rather than choose an algorithm from histogram appearance. Express effects in physical units and relate them to the robot task.

Deliverables: protocol.md, 30 raw-data files or one CSV with run_id, a summary table and a plot in test order. Edge cases: empty run, gaps, changed sampling rate and noticeable device motion. Define exclusion criteria before viewing results. Without independent ground truth, compare robustness and repeatability without claiming absolute IMU accuracy.

To introduce hypothesis testing, state H_0 : the mean paired difference between the two estimates is zero. Discuss false rejection of H_0 and missing a real effect; statistical significance does not say whether an effect is useful to the robot. Report the difference in physical units and its interval in the main conclusion, rather than a p-value alone.

Calibration optimisation: from error to parameters

Minimum theory

A loss function answers a specific question: how far do model predictions differ from measurements? Parameters θ may include encoder scale and zero; constraints express meaningful admissible values. The gradient gives the local direction of increasing loss; the Hessian describes local curvature. Begin with a two-variable quadratic bowl that can be drawn and whose derivatives can be checked by hand.

Gradient descent updates $\theta_{\text{new}} = \theta - \alpha \cdot \text{grad}$. Excessive α can increase error; too small an α slows progress. Learn backtracking line search: reduce the step until the objective decreases by the required amount. Distinguish stopping criteria based on step, gradient and loss; a small step may indicate poor scaling rather than a good minimum.

Task 7. Implement descent for scale and zero

Using M02 calibration, scale u to a comparable magnitude and derive the sum-of-squares gradient with respect to a and b . Check it with central finite differences at five parameter points. Run descent from three initial guesses with two step strategies; save the loss curve, iteration count and final coefficients.

On noise-free data, compare with linear `lstsq`. For poorly scaled u , explain elongated contours and slow convergence. Deliverables: short implementation, contour plot and table. Edge case: wrong gradient sign; a test must detect increasing loss and stop the run. Choose the gradient-check threshold relative to function scale and finite-difference step.

Task 8. Constrained nonlinear model

Use the teaching model $q = a \cdot u + b + c \cdot u^2$ with normalised u . Prepare independent calibration and verification series; generate a small known nonlinear component. Pass the residuals vector, not total loss, to `scipy.optimize.least_squares`. Specify bounds, initial guess and parameter scales; compare ordinary least squares and robust loss with labelled outliers.

Regularise c by adding residual $\sqrt{\lambda} \cdot c$ after choosing a dimensionless scale. Compare several λ values on the validation set, then evaluate the result once on the test set. Show a case where the linear model performs better on held-out data despite greater training error. The optimum is an estimate within the chosen model; still check residual dependence on position and motion direction.

SVD, pseudoinverses and uncertainty propagation

Core concepts

SVD factors $A=U \cdot \Sigma \cdot V^T$ into orthogonal input/output directions and stretch factors. A very small singular value indicates a parameter direction weakly represented in the data. A pseudoinverse selects a solution of a singular problem according to a defined rule; it cannot recover information the experiment never measured.

For $p=f(x)$ and small uncertainties near $\bar{x}$, use $\Sigma_p \approx J \cdot \Sigma_x \cdot J^T$. Σ_x is input covariance; evaluate J specifically at $\bar{x}$. Covariance has squared units of the corresponding output quantities. This is a local approximation, so compare with nonlinear Monte Carlo calculation, especially near geometric singularities or with large variability.

Task 9. Weakly determined calibration

Take the calibration matrix for a wide u range and the same matrix with inputs restricted to a narrow band. After applying the same scaling, compare singular values, cond and parameter changes under a small q perturbation. Find the right singular vector for the smallest singular value: it identifies the parameter combination least distinguishable in the data.

Save a plot of the uncertain parameter direction and predict which u values a new experiment should add. Then add extreme positions to the synthetic data and verify improvement. Test: all u values identical; pinv may return a solution, but the report must identify insufficient rank and nonunique physical calibration.

Task 10. Foot-position variability

Use $L=0.12$ m, $q=30^\circ$ and teaching standard uncertainties $\sigma_L=0.0005$ m and $\sigma_q=0.005$ rad. Repeat with large noise $\sigma_q=0.3$ rad; set nonzero correlation $\rho=0.6$ while keeping covariance positive semidefinite. Let $x=(L,q)$, and derive J for $p=(L \cos q, L \sin q)$. First assume independent errors, then add admissible nonzero covariance. Compute Σ_p , its eigenvector directions and a 10 000-point Monte Carlo cloud.

Compare coordinate SDs and covariance from the linear prediction with sample estimates. Repeat for small and large angle noise; record relative discrepancies while avoiding division by nearly zero covariance. The ellipse visualises the assumed uncertainty model, not a guaranteed real-robot position region. Negative sampled length requires revising the distribution model rather than silently deleting inconvenient points.

Integrated laboratory and six validity checks

Assemble the final package

Combine calibration and uncertainty: input `encoder_count` passes through the chosen $q(u, \theta)$ model, then geometry $p(L, q)$. For each validation run, save predicted position, estimated variability and actual error if independent ground truth exists. Synthetic experiments have known ground truth; the real stationary-IMU experiment usually establishes only repeatability.

Create three scenarios: nominal noise, correlated drift and outliers. Split independent series into training (train), validation and test sets. Choose the model, λ and threshold using train and validation; open test only after fixing settings. If you subsequently change the solution, mark the test set as used and prepare a new independent check: repeated tuning removes test independence.

Required error checks

1. Leakage. Compare random row splits with whole-run splits under slow drift; show why the former appears better. **2. Pseudoreplication.** Duplicate each row 100 times; the reporting algorithm must not count these as new independent experiments. **3. Outlier.** Add one large spike and compare mean, median and robust loss without deleting the original.

4. Singularity. Nearly constant u must trigger a weak-identifiability warning regardless of solver success. **5. Nonlinearity.** Large q noise invalidates linear covariance propagation; Monte Carlo must reveal the discrepancy. **6. Nonstationarity.** Different means at the start and end require a separate drift discussion rather than one 'sensor σ '.

Bootstrap: an additional interval check

Resample with replacement from the 30 independent run results and estimate intervals for the median or paired difference. Record resample count and seed; repeat with another seed to observe computational variation. Do not resample correlated individual rows as independent observations. Temporal dependence requires another scheme, such as block bootstrap, which is sufficient to identify as a later topic.

Distinguish observation and interpretation on every report page. A useful format is: 'between-run SD is ... under these conditions; it includes repositioning; no independent absolute reference is available'. Do not fill numerical fields in advance. Save metrics with units, interval bounds, independent-repeat count and specific applicability limits.

Session sequence: hours 1-40

Block A - build a noise model and identify its limits

Hours 1-2. Review error, true value and estimator definitions; prepare the data structure. **Hours 3-4.** Study distributions, expectation and variance; calculate probabilities of three discrete-encoder outcomes by hand. **Hours 5-6.** Study covariance and correlation with two channels. **Hours 7-8.** Study conditional probability and Bayes' rule through a diagnostic table. **Hours 9-10.** Study independence, temporal dependence and the experimental unit.

Hours 11-20 - practice. 2 h: quantisation and $\Delta/2$ check. 2 h: independent and shared IMU noise. 2 h: rare event, TP/FP/FN/TN. 2 h: white and correlated series. 2 h: plots, generator tests and conclusions. For each plot, state the sensor assumption built into the generator and a real effect it does not yet describe.

Hour 20 checkpoint: explain to someone without statistics training why high diagnostic sensitivity can coexist with many false alarms, and why 5000 adjacent samples do not equal 5000 repositionings. If you cannot, revisit the frequency table and temporal-dependence plot.

Block B - make a justified comparison

Hours 21-22. Study sampling and estimation; choose the measured quantity. **Hours 23-24.** Write the 30-repeat protocol and exclusion criteria. **Hours 25-26.** Study SD, standard error and a t-confidence interval for the mean. **Hours 27-28.** Study outliers, median, IQR and effect size. **Hours 29-30.** Study training, validation and test sets, paired comparison and repeatability.

Hours 31-40 - practice. 2 h: fixture preparation and pilot export; 2 h: 30 measurement runs with metadata; 2 h: integrity checks and individual-run summaries; 2 h: intervals and estimator comparison; 2 h: report and excluded-recording review. If longer recordings are necessary, reduce styling and use prepared summary scripts while retaining the independent-repeat count.

Checkpoint: another person can understand the dataset's experiment boundaries, what remained stationary, the coordinate frame used and why some recordings are marked doubtful. Without this information, measurements cannot support rigorous calibration comparisons in later months.

Session sequence: hours 41-75

Block C - optimise and verify calibration

Hours 41-42. Write the loss function for $q=a \cdot u+b$ and check dimensions. **Hours 43-44.** Derive and numerically check the gradient; draw curvature in two parameters. **Hours 45-46.** Study fixed steps and backtracking. **Hours 47-48.** Study nonlinear-model residuals, constraints and initial guesses. **Hours 49-50.** Study regularisation and model selection using validation data.

Hours 51-60 - practice. 2 h: implement descent and compare with `lstsq`; 2 h: `least_squares` with bounds; 2 h: outliers and robust losses; 2 h: several λ values and independent series; 2 h: gradient checks, model saving and residual analysis. For every run, retain status, evaluation count, initial guess, objective value and physical coefficients.

In the last two-hour segment, deliberately create a problem with local minima, such as periodic zero-offset ambiguity in $\sin(u+b)$, and start from different b values. This briefly introduces local minima and period ambiguity; do not add this complexity to the main calibrator. Conclude that results depend on the model and observational information, not merely the algorithm's name.

Block D - uncertainty in the result

Hours 61-62. Study SVD geometry and truncation of small singular values; check a diagonal example by hand. **Hours 63-64.** Derive J with respect to L and q , and propagate Σ . **Hour 65.** Study Monte Carlo, bootstrap, and measurement uncertainty versus computational variation in an estimate.

Hours 66-75 - practice. 2 h: poor calibration, SVD and a revised measurement plan. 2 h: analytical endpoint covariance. 2 h: Monte Carlo and large noise. 2 h: bootstrap by run and consolidated report. 2 h: six error checks, a fresh run and defending conclusions.

Final half-hour task without copying: specify two noise models with equal SD but different cross-axis dependence. Before running, predict which gives greater variation in a chosen foot-position projection. Check using the covariance formula and simulation. If the prediction was wrong, correct the explanation and preserve the mistake as a useful finding.

Do not measure success by method count. The required minimum is one explainable calibrator, one real experiment with justified conclusions, one poor-identifiability example and one error-propagation check. Rust/C++ may read M01 logs, but reimplementing all statistics in three languages is outside the 75 h.

Laboratory acceptance and the future robot

Verifiable deliverables

The package must include ten tasks with initial conditions, a 30-repeat protocol, labelled real and synthetic data, distribution models with limitations, calibration configurations, splits by series, residual plots, SVD diagnostics, covariance and Monte Carlo calculations. Record commands, library versions and report-reproduction order in README.

Every substantial number must answer: 'which data?', 'what unit of repetition?', 'what physical units?' and 'which metric?'. A mean without run count or an interval without a calculation method fails acceptance. Do not report more significant figures than uncertainty supports; preserve full machine precision separately for reproduction.

Check your understanding

Explain variance versus SD; independence versus zero correlation; random spread versus systematic bias; repeatability versus absolute accuracy. Use a table to distinguish $P(\text{failure}|\text{alarm})$ from $P(\text{alarm}|\text{failure})$. Why can you not narrow a confidence interval simply by copying rows?

For optimisation: what goes into residuals; why normalise parameters; why is success insufficient; what does a small singular value reveal; what does regularisation change, and what does it not know about the robot? For uncertainty: where is J evaluated; what are Σ 's units; when does linear propagation fail; which units should bootstrap resample?

Practical assessment: take a new independent synthetic series with a seed unknown to you, calibrate and compare estimates with saved ground truth. Add drift and state which assumptions fail. For real data, show one supported conclusion and one requiring a different reference or experiment.

Handover

Pass M04 the actual sampling rates, gaps, noise summaries, original recordings and corrupted copies for filtering work. Pass M05 length-and-angle error propagation as an example connecting tolerances to foot position. Later filters and multi-joint calibration can reuse these functions, but a sensor's numerical covariance must be remeasured after mounting or settings change.

Keep open questions about temperature drift, vibration dependence, axis correlation, reverse mechanism travel and reference accuracy. For each, describe the smallest future experiment that could test it. This makes the laboratory a foundation for later hardware tests, beyond a collection of one-off plots.

Assigned reading: a short route

Read only the specified passages before the corresponding experiment. References are mainly in English; keep explanations, glossary and report in English. Reading and preparation fit within the existing 75 hours; whole books are not required.

Core reading

1. Allen B. Downey, Think Stats, 3rd online edition. [Chapters 2, 4, 7, 8](#): Distributions; Cumulative Distribution Functions; Relationships between variables; Estimation. Tasks 1-6 need distributions, relationships and estimation uncertainty; adapt the examples to your measurements.

2. NIST/SEMATECH, e-Handbook of Statistical Methods. [§1.3.5.2 Confidence Limits for the Mean](#): interval from 30 independent runs; [§1.3.5.11, skewness and kurtosis](#): noise-shape reference. Write interval assumptions before inserting numbers.

3. Stephen Boyd, Lieven Vandenberghe, Convex Optimization (2004). [§4.1, §6.1, §6.3, §9.2-9.3](#). Problem formulation; norm approximation; regularisation; descent and gradient methods. In tasks 7-8, connect loss, constraints and parameter scale to physical calibration. General-case proofs are optional.

Compact implementation references

SciPy optimize. [least_squares: fun, jac, bounds, x_scale, loss](#). Before task 8, check stopping criteria and output fields. A success status does not replace unit, bound and held-out-data checks.

NumPy. [svd: Returns, Notes](#). For task 9, reconstruct A from U,S,Vh, then compare the pseudoinverse with pinv. Explain what changes when a small singular value is truncated.

SciPy stats. [bootstrap: data, statistic, paired, method](#). Resample independent runs; use paired only for paired comparisons. Optional constrained least-squares comparison: [CVXPY example](#).

Reading sequence. First distributions and the 30-repeat plan, then estimates and intervals, then optimisation and SVD. Deliver one sheet of experiment assumptions and one defining the calibration problem.

Glossary: terms and definitions

Random variable. A numerical function of a possible random-experiment outcome; its distribution specifies probabilities of values or ranges.

Independence. Property of events or variables whereby information about one does not change the probability distribution of the other.

Conditional probability. Probability of A given B: $P(A|B)=P(A \cap B)/P(B)$, provided $P(B)>0$.

Sample. The particular set of observations obtained; under temporal dependence, its size is not the number of independent observations.

Parameter and estimate. A parameter characterises an unknown model; an estimate approximates it from a particular sample using a specified rule.

Bias. Difference between an estimator's expectation and the true parameter. Averaging does not remove constant sensor bias.

Variance. Expected squared deviation from the mean; describes variability and has squared units of the measured quantity.

Standard deviation. Square root of variance; describes individual-value variability in original units, such as rad or m/s².

Standard error. Standard deviation of an estimator's distribution over repeated samples; describes estimation uncertainty rather than individual-reading variability.

Covariance. Mean product of centred deviations of two quantities; its sign describes the direction of their joint linear variation.

Correlation. Normalised covariance; describes linear association, but zero generally does not imply independence.

Confidence interval. Interval produced by a procedure that, over repeated experiments, covers a fixed parameter at the specified frequency.

Bootstrap. Estimation of a statistic's distribution by repeated sampling with replacement. The resampling unit must match the original experiment's independent unit.

Loss function. Scalar measure of model-data discrepancy; its choice determines which errors have greater influence on calibration.

Gradient. Vector of a scalar function's partial derivatives; gives the local direction of greatest increase for the chosen coordinates and metric.

Regularisation. An additional preference or penalty in estimation that limits unstable solutions at the cost of possible systematic bias.

SVD (singular value decomposition). Factorisation $A=U\Sigma V^T$ for a real matrix; singular values describe amplification along mutually orthogonal directions.

Identifiability. Ability to distinguish parameters from available observations and inputs; repeating the same experiment can preserve model ambiguity.

Equipment, installation and checks

Complete preparation within the first study blocks in place of another general overview. The budget remains 75 hours; delivery waiting time is excluded. Statuses below specify actions for you to take, not purchases or installations already completed.

Use existing equipment

Use the MacBook Pro M4 and M02 Python environment with pinned NumPy, SciPy, Matplotlib, pytest and the chosen notebook environment. For physical work, use **an existing phone with raw IMU data or an existing IMU**, a stable fixture and a way to repeat placement. Perform 30 separate placements and runs, not 30 adjacent samples.

Purchases if needed

No new mandatory M03 purchases. A motor and separate new IMU board are unnecessary before M06. If the phone cannot supply the required data, use an available existing sensor; synthetic work can begin immediately, but the real experiment remains open until 30 physical repeats are completed.

Install now

On the phone: an app exporting original accelerometer readings, timestamps and units; for example, [phyphox](#), [Acceleration with g](#). Check sensor availability on your model. Choose readings including gravity; do not describe 'linear acceleration' after hidden processing as a raw channel.

On macOS ARM: no new mandatory packages; use `scipy.stats` and `scipy.optimize` from M02. If the file format requires another package, add only that package and update the lockfile. Install CVXPY solely for the optional constrained-calibration comparison, not as an M03 passing requirement.

Readiness checks

Before the series, export one pilot CSV containing time in seconds, three axes, units, run identifier and sensor-mode name. Check finite values, timestamp ordering, gaps and duration. On a stationary support, raw accelerometer magnitude should be near g; this checks mode and scale, not full calibration.

Open the CSV after restarting the notebook. On synthetic data with a known seed, calculate mean, SD and an interval; recover known a, b with `least_squares` and check SVD matrix reconstruction. Save software versions and the data manifest. Successful optimisation does not replace checking ground truth and run independence.

Before the next stage

The raw-data archive must contain 30 identifiable physical repeats or explicitly mark that part incomplete. Keep export settings and synthetic data separately; assess laboratory readiness against the original M03 criteria without adding hours.

Motion fundamentals: how angle error moves the foot

Compulsory, within 75 hours. Use 1 h from 63-64 and 4 h from block 66-75 already allocated to analytical covariance and Monte Carlo in task 10. No new measurements or equipment are required. Entry check: explain zero-offset error, random variability and standard error of the mean.

Displacement direction depends on the error source

Use the task 10 link: $L=0.12$ m, $q=\pi/6$ rad, in a fixed planar frame with x right and y upwards; $p=(L \cos q, L \sin q)$. Parameters $u=(L, q)$ have small independent zero-mean errors: $\sigma_L=0.0005$ m, $\sigma_q=0.005$ rad. This is a specified teaching model, not a claim about real-encoder accuracy.

A small length error moves the endpoint along the link: $\partial p / \partial L = (\cos q, \sin q)^T$. Angle error acts tangentially: $\partial p / \partial q = (-L \sin q, L \cos q)^T$. Assemble these columns into J to obtain $\delta p \approx J \delta u$. Here u denotes the parameter pair, not `encoder_count` in the calibration model.

Covariance transforms as $\Sigma_p \approx J \cdot \text{diag}(\sigma_L^2, \sigma_q^2) \cdot J^T$. Independent input errors do not imply independent x and y : both outputs depend on L and q . Standard spread along the link is 0.5 mm and across it $L \sigma_q = 0.6$ mm; these are ellipse semiaxes at one standard uncertainty, not a guaranteed foot-position region.

A calculation you can check by hand

At 30° , $\Sigma_{xx}=0.2775$ mm², $\Sigma_{yy}=0.3325$ mm², $\Sigma_{xy}=\Sigma_{yx}=-0.047631$ mm². Thus $\sigma_x \approx 0.5268$ mm and $\sigma_y \approx 0.5766$ mm. The matrix trace is 0.61 mm², the sum $0.5^2 + 0.6^2$; rotating axes changes components but preserves this trace.

Now add a constant zero-offset error $\Delta q = +0.01$ rad. The linear mean shift $J_q \Delta q \approx (-0.600; 1.039)$ mm has length 1.2 mm. The exact change $p(q + \Delta q) - p(q) \approx (-0.605; 1.036)$ mm. Repeating one position can reduce uncertainty in the estimated mean, but does not remove this constant geometric shift.

Verify task 10 and decide what to calibrate

1. Using the same 10 000-point Monte Carlo cloud, compare both variances and covariance with the calculation, then rotate axes by 90° and check trace and eigenvalues. Save a drawing of the link, ellipse and cloud. Do not use relative error of nearly zero covariance as the sole metric; allow for Monte Carlo's own variability.

2. Add Δq separately and show the cloud-centre shift; then restore the task's large $\sigma_q = 0.3$ rad. Explain why a tangent describes the arc less accurately. In the report, state what needs correcting: zero offset, length or random-error model. M03 phone repeatability alone does not calibrate a future leg's geometry.

Read before calculating: NIST/SEMATECH, [§2.5.5: propagation of error, sensitivity coefficients](#); Lynch, Park, [Modern Robotics, chapter 5 introduction: Jacobian](#).